Audiveris sandbox build & run notes

1. Rebuild in a new session

```
git checkout claude/audiveris-sandbox-build
dev/sandbox/setup.sh                # ~3-5 min the first time
dev/sandbox/run.sh my.pdf out 2 5-15 # ~5 min for 12 pages
```

2. Environment blockers & workarounds

Blocker	Symptom	Workaround
JDK 21 preinstalled	`invalid source release: 25`	`apt install openjdk-25-jdk-headless`
repository.jboss.org blocked	`Could not resolve javax.media:jai-core:1.1.3` (403)	Empty stub jar + pom in `~/.m2` (unused by code)
Maven Central rate limit	HTTP 429	Retry loop in `setup.sh`
Ubuntu tessdata is LSTM-only	`Could not initialize TessBaseAPI ... legacy mode`	Download full `eng.traineddata` from tesseract-ocr/tessdata

3. Audiveris bugs fixed (commit “Fix batch -print and multi-movement export with -sheets”)

1. **Batch `-print` crashed** (`ExceptionInInitializerError: Application is not launched`). `SheetPainter` and `ViewParameters` read GUI resource maps in static initializers. Fix: resolve them lazily, only where the GUI needs them.
2. **Empty (0 byte) print PDF** when any sheet painting failed. The exception skipped `g2.dispose()`, and iText then threw `Unbalanced save/restore state operators` on close. Fix: per-sheet try/catch/finally in `BookPdfOutput.write`.
3. **MusicXML movements overwrote each other** as `<book>.mvtnull.mxl` when using `-sheets`. Scores built from selected sheets aren’t registered in the book, so `Score.getId()` was null. Fix: `Book.getMovementId` falls back to the score rank in the exported list.

4. Open issues (not fixed)

- Sheets that throw during OMR abort the whole batch (`Error in reaching step PAGE`). Workaround: exclude them with `-sheets`.
- Observed NPEs on the test score:
  - `stems is null` (sheet 3)
  - `Measure.getClefBefore ... measure is null` (sheet 4)
  - `Staff.getPart() ... this.staff is null` in `SymbolsBuilder` (sheet 8, non-fatal)
  - `PartBarline with no proper StaffBarline` (non-fatal, measures dropped from export)

5. First test run: “Incident at Wounded Knee”, Bassoon II (15-page scan)

- Command: `run.sh Bassoon_II_-_Ballard.pdf out 2 5-15`
- Sheet 1: cover page (no staves). Sheets 3-4: crash (see above).
- Output: 12-page engraved PDF, 7 MusicXML movements, 422 measures total.
- Quality is rough, so expect manual correction in the Audiveris GUI:
  - header clefs mostly missed, so pitches and accidentals are unreliable
  - time signatures missed, so measure durations aren’t checked
  - multi-measure rests and some barlines misread
  - the “RENTAL ONLY” stamp gets OCR’d as garbage text
- Tips to improve: rescan at 300-400 dpi, clean stamps/marks, then fix clefs and time signatures in the GUI and re-run from the `.omr` project.

6. Output structure (audiveris-results.zip)

```
audiveris-results/Bassoon_II_-_Ballard-print.pdf  (420 KB)
audiveris-results/Bassoon_II_-_Ballard.mvt1.mxl  (4 KB)
audiveris-results/Bassoon_II_-_Ballard.mvt2.mxl  (4 KB)
audiveris-results/Bassoon_II_-_Ballard.mvt3.mxl  (8 KB)
audiveris-results/Bassoon_II_-_Ballard.mvt4.mxl  (8 KB)
audiveris-results/Bassoon_II_-_Ballard.mvt5.mxl  (20 KB)
audiveris-results/Bassoon_II_-_Ballard.mvt6.mxl  (4 KB)
audiveris-results/Bassoon_II_-_Ballard.mvt7.mxl  (4 KB)
audiveris-results/Bassoon_II_-_Ballard.omr  (2920 KB)
audiveris-results/run.log  (164 KB)
```

File	What it is
`*-print.pdf`	Engraved score, 12 pages (sheets 2, 5-15)
`*.mvt1..7.mxl`	Compressed MusicXML, one per movement; open in MuseScore/Dorico/Sibelius
`*.omr`	Audiveris project; open in Audiveris GUI to correct and re-export
`run.log`	Full processing log

7. Git

- `claude/audiveris-pdf-engraving-c59j80`: bug-fix commit `decf2c5`
- `claude/audiveris-sandbox-build`: `decf2c5` + scripts/notes/CLAUDE.md `f7d5b5f`
- If not pushed: `git fetch audiveris-branches.bundle "refs/heads/*:refs/heads/*"` then `git push origin claude/audiveris-sandbox-build`